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B.Tech. Degree IV Semester Examination April 2018

ME 15-1402 METROLOGY AND INSTRUMENTATION (2015 Scheme)

Time : 3 Hours

Maximum Marks : 60

PART A (Answer *ALL* questions)

(10 × 2 = 20)

- I. (a) Mention the difference between line standard and end standard of measurement.
 (b) Differentiate the following.
 (i) Hole basis system and shaft basis system (ii) Tolerance and allowance
 (c) Explain "Parallax error" with a suitable sketch.
 (d) In the measurement of surface roughness, heights of successive peaks and troughs were measured from a datum and were 33, 25, 19, 22, 18, 27, 29 and 20 microns. If these measurements were obtained on 10 mm length, determine CLA and RMS values of surface roughness.
 (e) What is an optical flat? State the limitations of optical flats.
 (f) Distinguish between active and passive transducers.
 (g) What is static calibration? Explain.
 (h) Describe the working of potentiometer transducer as a zero order instrument.
 (i) Define load cells. What are the types of load cells?
 (j) Explain gauge factor.



PART B

(4 × 10 = 40)

- II. In a hole and shaft assembly of 30 mm nominal size, the tolerances for hole and shaft are as specified below: (10)
 Hole: $30_{-0.000}^{+0.02}$ mm Shaft: $30_{-0.070}^{-0.040}$ mm
 Determine:
 (i) Maximum and minimum clearance obtainable
 (ii) Allowance
 (iii) Hole and shaft tolerance
 (iv) MML of shaft and hole.
 (v) The type of fit.

OR

- III. (a) Discuss about the importance of gauging in manufacturing process. With neat sketches explain the different classification of gauges. (6)
 (b) Explain what is meant by interchangeability and selective assembly. (4)
- IV. Explain how sine bar is used to measure:
 (a) Angle of a component of small size. (7)
 (b) List the possible sources of errors in angular measurement by sine bar. (3)

OR**(P.T.O.)**

- V. (a) Explain the working of NPL flatness type interferometer with the help of a neat sketch. (8)
(b) Explain why monochromatic light is used for interferometry work and not the white light. (2)
- VI. (a) What are the main functional elements of an instrument? Mention the functions of each element. (6)
(b) Define the following terms. (4)
(i) Sensitivity.
(ii) Hysteresis.
- OR**
- VII. What are the different methods of correcting spurious inputs in an instrument? Explain with examples. (10)
- VIII. Explain the methods used for temperature compensation in strain gauges. (10)
- OR**
- IX. (a) With a neat sketch explain Heenan and Froude hydraulic dynamometer. (6)
(b) Explain the working of an optical pyrometer. (4)
